Stochastic Harmonic Hosting Capacity Considering Harmonic Recombination

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***Abstract*—Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.**

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***Index Terms*—Harmonic distortion, fairness, hosting capacity, mathematical optimization.**

I. Introduction

A. Background and Motivation

REACHING a net-zero economy by 2050 requires extensive electrification of the energy system. Over the past two decades, renewable generation—particularly wind and photovoltaic (PV) power—has grown steadily and is expected to accelerate towards 2050 [**?**]. At the same time, decarbonization targets necessitate the electrification of transport, heating, and industrial processes, replacing fossil fuels with electricity. In Europe, studies project that total electricity demand will triple by 2050 [**?**]. Consequently, future power systems will be largely dominated by power-electronic interfaced resources on both the supply and demand sides. This transition fundamentally changes the harmonic behavior of power systems. Traditional grids, dominated by synchronous machines and mostly linear loads, exhibited limited harmonic distortion. Modern systems, however, increasingly rely on converters whose nonlinear characteristics generate a broad spectrum of harmonic currents. The aggregation of numerous distributed converters creates new challenges in maintaining harmonic voltage quality across transmission and distribution networks.

A key challenge arises from the stochastic nature of harmonic emissions. Measurements and laboratory studies show that harmonic magnitudes depend on loading, switching states, and internal control actions. More importantly, harmonic current phase angles—which govern the phasor summation of emissions from multiple sources—vary significantly. Even small angle variations can result in large differences in harmonic voltage distortion, particularly in networks with many distributed converters. As converter penetration increases, system operators need transparent metrics to assess whether additional units can be connected without exceeding harmonic limits. Harmonic hosting capacity provides such a metric by defining the maximum allowable harmonic emissions while ensuring compliance with power-quality standards. However, most existing methods rely on deterministic assumptions regarding harmonic magnitudes, phase angles, network parameters, and background distortion [**?**], [**?**], leading to either overly conservative or insufficiently protective results.

This work presents a harmonic hosting capacity methodology that explicitly accounts for the stochastic nature of harmonic current phase angles associated with power-electronic interfaced resources, thereby capturing harmonic recombination effects. This enables the enforcement of stochastic limits on harmonic voltages and currents in line with relevant standards.

B. Literature Review

There are several standards (std.) and, more recently, limited academic literature (lit.) that present an approach to determine the harmonic hosting capacity of a power system. As highlighted in § I.B Literature Study, Table I [**?**], there is still a significant gap with respect to stochastic representation in the context of harmonic hosting capacity. Table I augments the aforementioned table, extending it with relevant work discussing stochastic representation, including on: (a) unit harmonic injection, (b) background harmonic distortion, (c) network parameters, (d) network exploitation, and (e) harmonic limits.

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TABLE I
LITERATURE REVIEW ON HARMONIC HOSTING CAPACITY, ENUMERATING FEATURES WITH RESPECT TO THE CATEGORIES ESTABLISHED IN § I-A. THE LAST LINE OF THE TABLE SHOWS THE FEATURES INCLUDED IN THE PRESENTED PAPER.

ref.	class	objective	method[1]	ntw. repr.		stoch. method[2]	stoch. repr. (a)	(b)	(c)	(d)	(e)	tract.[3] #n	t[min]
[?]	std.	harmonic hosting capacity	ae	nodal	bal.	-	✗	✗	✗	✗	✗	-	-
[?]	std.	harmonic hosting capacity	ae	nodal	bal.	-	✗	✗	✗	✗	✗	-	-
[?]	std.	harmonic hosting capacity	ae	nodal	bal.		✗	✗	✗	✗	✗	-	-
[?]	lit.		ae	nodal	bal.	-	✗	✗	✗	✗	✗	14	-
[?]	lit.		ae	nodal	bal.		✓	✗	✗	✗	✓	64	-
[?]	lit.		pf	full	unb.	-	✗	✗	✗	✗	✗	8773	320
[?]	lit.		pf	full	bal.		✓	✓	✗	✗	✗	3	-
[?]	lit.		nl	full	bal.		✓	✗	✗	✗	✗	16	-
[?]	lit.		nl	full	bal.		✓	✗	✗	✗	✗	5	-
[?]	lit.		nl	full	bal.		✓	✗	✗	✗	✓	16	30
[?]	lit.		soc	full	bal.	-	✗	✗	✗	✗	✗	1880	< 6
This work	lit.		soc	full	bal.	pce	✓	✗	✗	✗	✓	1880	< 6

1) Stochastic modeling of harmonic injections: Early stochastic harmonic studies focused on probabilistic aggregation of harmonic sources. Emanuel and Pileggi modeled harmonic currents as random variables to capture diversity effects, thereby addressing uncertainty in unit harmonic injection, corresponding to category 4a in Table I [**?**]. Bollen and Gu further demonstrated that random phase angles significantly reduce expected harmonic voltages compared to deterministic in phase assumptions, highlighting the importance of phase angle uncertainty, which is not explicitly represented in Table I [**?**].

Monte Carlo based harmonic power flow methods were later introduced to simultaneously capture uncertainty in harmonic injections and network parameters, addressing categories 4a and 4c. Xu et al. incorporated stochastic harmonic current magnitudes and impedance variations [**?**], while Langella et al. derived probability distributions for harmonic injections based on field measurements of nonlinear loads [**?**]. Although these approaches improve realism, their reliance on repeated simulations limits tractability for large scale hosting capacity studies.

2) Stochastic harmonic hosting capacity: The explicit treatment of uncertainty in harmonic hosting capacity is comparatively recent. Sun and Harrison proposed a chance constrained optimal power flow formulation that incorporates probabilistic harmonic voltage distortion limits, thereby addressing uncertainty in background harmonic distortion and harmonic limits, corresponding to categories 4b and 4e in Table I [**?**]. Their results show that probabilistic constraints can significantly increase admissible hosting capacity compared to deterministic limits.

Nakhodchi and Bollen extended Monte Carlo based hosting capacity analysis to electric vehicle fast charging stations, modeling stochastic harmonic injections and implicitly accounting for phase angle diversity [**?**]. Their study demonstrates that deterministic worst case assumptions may substantially underestimate hosting capacity. However, the approach remains computationally intensive and does not yield guaranteed worst case bounds.

3) Uncertainty in harmonic current phase angles: Uncertainty in the phase angle of injected harmonic currents is particularly influential, as harmonic voltages arise from vector summation across multiple sources. Measurement based studies consistently report wide dispersion of harmonic current angles even among similar devices [**?**], [**?**]. Analytical studies by Wang et al. showed that assuming random phase angles leads to closed form expressions for expected harmonic voltages that are significantly lower than deterministic worst case estimates [**?**].

Polynomial chaos expansion has been proposed as an efficient alternative to Monte Carlo simulation for propagating uncertainty in both harmonic current magnitude and phase. Zaninelli and Sulis demonstrated that this approach can accurately represent stochastic harmonic power flows while remaining computationally efficient [**?**]. These methods directly address uncertainty in harmonic current angles, which is not explicitly represented as a separate category in Table I.

From a hosting capacity perspective, Salles et al. quantified the impact of harmonic phase diversity on admissible converter penetration levels, concluding that deterministic in phase assumptions represent an extreme lower bound scenario [**?**]. This observation directly motivates the interpretation of deterministic harmonic hosting capacity as a conservative lower bound, as adopted in the present paper.

4) Mapping to Table I and identified gaps: Table I shows that existing literature partially addresses stochasticity through Monte Carlo simulation and chance constrained formulations, covering categories 4a, 4b, and 4e. However, no tractable optimization based harmonic hosting capacity method explicitly models uncertainty in harmonic current phase angles. Furthermore, uncertainty in network exploitation and topology changes remains largely unexplored.

To better reflect the state of the art, an additional conceptual row could be added to Table I, representing stochastic phase angle modeling. Such a row would be characterized by probabilistic or parametric representations of harmonic current angles, potentially using analytical summation, polynomial chaos expansion, or scenario based techniques. Existing

[1]The possible mathematical methods are abbreviated as: (ae) analytical expression, (pf) power flow, (nl) non-linear optimal power flow, and (soc) second-order cone optimal power flow.

[2]The possible stochastic methods are abbreviated as: (mc) monte-carlo analysis, (ft) field-theory, and (pce) polynomial chaos expansion.

[3]The tractability of a method is expressed in terms of size of the network studied, i.e., number of nodes #n, and the solve time t given in minutes.

studies addressing this aspect improve realism but do not yet provide tractable worst case guarantees suitable for large scale planning.

In this context, the deterministic assumption of fixed harmonic current angles provides a device agnostic and worst case compliant definition of harmonic hosting capacity. At the same time, the reviewed literature indicates that incorporating stochastic phase angle representations could substantially relax this lower bound. Bridging this gap between tractability and stochastic realism remains an open research direction.

C. Contributions and Outline

In this paper, a holistic one-shot methodology is presented to exactly and tractably determine the steady-state harmonic hosting capacity for an entire phase-balanced three-phase power system in a stochastic context. The presented method considers uncertainty on the angle of the injected harmonic current, enabling harmonic recombination, and limits the resulting individual and total harmonic voltage distortion, as well as the root-mean-square voltages and current, using chance constraints. The problem is formulated from the perspective of the power system operator, i.e., without explicit knowledge of the (future) assets of its costumers. Consequently, the harmonic hosting capacity is determined in terms of harmonic current magnitude using a fairness principle. The presented formulation using the rectangular current-voltage variable space is second-order conic, incorporating:

- uncertainty on the angle of the injected harmonic current,
- exact representation of the network impedance in the frequency domain based on the harmonic impedance of the individual grid components,
- detailed treatment of transformers including the effects of their configuration,
- enforcement of stochastic (harmonic) limits given in relevant standards on bus voltage and component ampacity, and
- equitable distribution of the harmonic hosting capacity following a fairness principle.

The models and numerical illustrations presented in the paper are made freely available as part of the JULIA-package: HARMONICPOWERMODELS.JL[4].

The remainder of the paper is organized as follows: Section II addresses the uncertainty of harmonic current phasors associated with power-electronic interfaced resources. Section III presents the mathematical formulation of the stochastic harmonic hosting capacity problem. Section IV shows numerical illustration results for a radial section of a real-world industrial power system and for a large-scale transmission system. Section V concludes the paper.

D. Literature Review

Harmonic hosting capacity analysis has historically relied on deterministic harmonic models and worst case assumptions. However, stochastic variability in nonlinear injection sources, network conditions, and background distortion motivates probabilistic representations to more accurately capture harmonic impacts on power quality compliance. Recent studies have begun to incorporate stochastic descriptions into harmonic modeling and hosting capacity, addressing aspects of category 4 in Table I related to uncertainty representation.

1) Stochastic harmonic power flow and uncertainty propagation: Several researchers have developed probabilistic harmonic load flow (PHLF) methods that model uncertainties in harmonic injections and network responses. Xie et al. propose a data driven probabilistic harmonic power flow in distribution systems with photovoltaic generation, where harmonic sources are characterized as random processes and Monte Carlo sampling methods are used to obtain probability distributions of harmonic voltages and currents [**?**]. Rodríguez-Pajarón et al. present a piecewise probabilistic harmonic power flow approach that represents residential load harmonics and network uncertainty through statistical distributions, demonstrating improved accuracy and reduced computational burden compared to brute force sampling [**?**]. Both works contribute to categories 4a and 4c, as they address stochastic harmonic sources and their effects on voltage distortion.

Cadena-Zarate et al. study probabilistic harmonic load flows for distribution networks with distributed energy resources, capturing random behavior of harmonic injections from inverter-based sources through simulation based methods [**?**]. These probabilistic load flow frameworks quantify the variability of harmonic distortion metrics and illustrate significant differences from deterministic predictions.

Li et al. provide a formal stochastic assessment of harmonic propagation and amplification in power systems under uncertainty, categorizing system uncertainties and using Monte Carlo techniques to characterize their influence on harmonic interactions and amplification phenomena [**?**]. This work helps inform the class of stochastic representations relevant to propagation and network response uncertainties (categories 4c and 4f).

2) Stochastic hosting capacity formulations: Hosting capacity assessments that incorporate harmonic uncertainty remain scarce but important. Sun and Harrison formally integrate probabilistic harmonic distortion limits into an optimal power flow based hosting capacity model using chance constrained formulations [**?**]. Their approach models uncertainty in background harmonic distortion and load, corresponding to categories 4b and 4e, and demonstrates that relaxing tight risk constraints can increase permissible injection capacity.

In addition, probabilistic hosting capacity enhancements have been proposed for non-sinusoidal systems using Monte Carlo based optimization, showing how various uncertain parameters such as generation output and background distortion affect harmonic constrained hosting capacity [**?**]. Although not specific to phasor angle uncertainty, such frameworks illustrate how uncertainty at different model levels influences harmonic limits.

3) Uncertainty in harmonic current phase angles: The phase angles of injected harmonic currents critically affect vector summation of harmonics at points of common coupling, yet few studies explicitly model their uncertainty within

[4]Available at: https://github.com/timmyfaraday/HarmonicPowerModels.jl

hosting capacity contexts. Arghandeh et al. introduce the Phasor Harmonic Index, which incorporates both magnitude and phase angle of harmonic currents to assess interactions among distributed energy resource injections [**?**]. While this work focuses on interaction indices rather than hosting capacity per se, it highlights the importance of phase angle in stochastic harmonic models.

Further, stochastic studies of harmonic configurations, such as those examining the effects of phase angle interval size on harmonic cancellation phenomena in wind farms, show that wider randomized angle distributions lead to lower expected total harmonic distortion due to cancellation effects [**?**]. This type of uncertainty characterization relates directly to category 4d and reveals that deterministic worst case alignments represent conservative extremes.

4) Mapping to Table I and research gaps: Existing probabilistic harmonic load flow literature addresses stochastic harmonic injections, background distortion, and network response uncertainties (categories 4a, 4c, and 4f). Hosting capacity research that incorporates probabilistic distortion limits through chance constraints contributes to categories 4b and 4e. However, *explicit modeling of harmonic current phase angle uncertainty in hosting capacity optimization frameworks* remains an open area. Additionally, categories relating to correlated uncertainties among multiple harmonic sources and network topology changes are largely unexplored in hosting capacity contexts.

To better reflect these developments, an additional conceptual row can be added to Table I for *stochastic phase angle models*. Such a row would include representations where harmonic current angles are described using probability distributions (e.g., uniform, empirically derived distributions) or parametric stochastic processes coupled with analytical or approximate harmonic summation techniques, enabling tractable assessment of expected distortion and hosting limits under angle uncertainty.

In summary, while deterministic harmonic hosting capacity provides a conservative lower bound, stochastic representations that include phase angle variability and other sources of uncertainty can offer less conservative and more realistic capacity estimates. Bridging tractable optimization formulations with comprehensive stochastic modeling represents a key research frontier.

II. Angle Uncertainty related to Converters

This section addresses the uncertainty of harmonic current phasors, especially phase angles, related to the typical industrial load with a 6-pulse line-commutated converter (LCC) as shown in Fig. 1. The modeling method integrates steady-state harmonic state-space (HSS) model into Monte Carlo simulation (MCS), which enables the assessment of uncertainty propagation from key circuit parameters to harmonic phase angle.

First, a steady-state HSS model of LCC is established, which captures the mutual coupling among the harmonic phasors and their dependence on key passive parameters, namely the commutation inductance L_{ac} on the AC side and the smoothing inductance L_{dc} on the DC side. Then, the parameterized model is integrated into a stochastic assessment framework. The key parameters are modeled as random variables with realistic bounds. The MCS is applied to genetate probability distributions of harmonic phasors. Finally, the marginal probability density functions (PDFs) are fitted using the best-fit method, and the Gaussian copula correlation matrix of harmonic phase angles is obtained to indicate the inter-harmonic correlation.

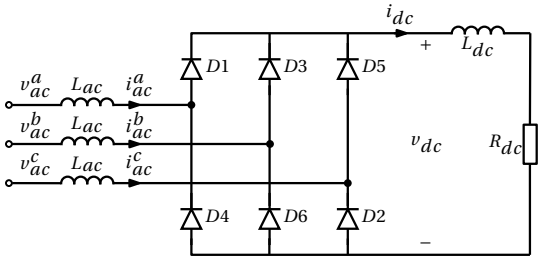


Fig. 1. 6-pulse line-commutated converter

A. Steady-State Harmonic Model

Inherent frequency coupling is the key feature of converter-generated harmonics, which decide the recombination of harmonic phasors. Therefore, the accurate HSS model is built here to capture the nature commutation and harmonic coupling of LCC converter.

The LCC converter operates periodically with sequential valve switching creating 12 switching states per fundamental cycle. Each state is governed by differential equations with time-varying coefficients representing circuit topology changes. The state equations for any switching state can be expressed using piecewise switching functions $S^i_{dc,ac}(t)$, $S_{dc}(t)$, and $S^{i,j}_{ac}(t)$ as:

$$\begin{aligned}&(S_{dc}L_{ac}+L_{dc})\frac{di_{dc}}{dt}=\\&S^a_{dc,ac}v^a_{ac}+S^b_{dc,ac}v^b_{ac}+S^c_{dc,ac}v^c_{ac}-R_{dc}i_{dc},\end{aligned}\tag{1}$$

$$\begin{aligned}&L_{ac}\frac{di^i_{ac}}{dt}-S^i_{dc,ac}L_{ac}\frac{di_{dc}}{dt}=\\&S^{i,a}_{ac}v^a_{ac}+S^{i,b}_{ac}v^b_{ac}+S^{i,c}_{ac}v^c_{ac},\end{aligned}\tag{2}$$

where v^i_{ac} and i^i_{ac} denote AC-side voltage and current, the superscript $i \in \{a,b,c\}$ represents the three phases, v_{dc} and $i_{dc}(t)$ are DC-side voltage and current. All these switching functions are parameterized by the commutation angle μ in steady state.

The HSS model transforms these time-domain equations into frequency domain through three operations: (1) Fourier series expansion of state variables $x(t)=\sum_{h=-N_h}^{N_h} X_h e^{jh\omega_1 t}$, (2) frequency-domain differentiation $d\mathbf{X}/dt \leftrightarrow \mathbf{Z}_\Omega\mathbf{X}$ where $\mathbf{Z}_\Omega = j\omega_1 \cdot \mathrm{diag}(h)$, and (3) Toeplitz convolution matrices representing time-domain multiplication: $y(t)=S(t)x(t) \Leftrightarrow \mathbf{Y}=\mathbf{S}_{conv}\mathbf{X}$. The resulting unified system relates current harmonics $\mathbf{I}=[\mathbf{I}^a_{ac},\mathbf{I}^b_{ac},\mathbf{I}^c_{ac}]^T$ to voltage harmonics $\mathbf{U}=[\mathbf{V}^a_{ac},\mathbf{V}^b_{ac},\mathbf{V}^c_{ac}]^T$ through:

$$\mathbf{AI}=\mathbf{BU},\tag{3}$$

where matrices $\mathbf{A}$ and $\mathbf{B}$ contain Toeplitz convolution structures explicitly capturing frequency coupling. The model is parameterized by commutation overlap angle μ, and circuit inductances (L_{ac}, L_{dc}), providing a foundation for subsequent MSC and uncertainty quantification.

B. Methodology for Uncertainty Quantification

At the planning stage, the uncertainty on key parameters comes from the limited knowledge about the future integrated device, as well as manufacturing tolerances and operational variances. To account for these parameter uncertainties, the inductances L_{ac} and L_{dc} are modeled as random variables, and the MCS is used for assessment of uncertainty propagation.

1) Realistic parameter uncertainty bounds: The uncertainties on L_{ac} and L_{dc} are modeled using truncated normal distributions:

$$L_{ac} \sim \mathcal{N}_{\text{trunc}}(\mu_{L_{ac}}, \sigma^2_{L_{ac}}, L_{ac,\min}, L_{ac,\max}) \tag{4}$$

$$L_{dc} \sim \mathcal{N}_{\text{trunc}}(\mu_{L_{dc}}, \sigma^2_{L_{dc}}, L_{dc,\min}, L_{dc,\max}) \tag{5}$$

which are constrained within physically realistic bounds determined by engineering trade-offs at the industrial load design stage:

- *AC-side inductance L_{ac}:* The upper bound $L_{ac,\max}$ is limited by acceptable commutation angle and voltage drop, while the lower bound $L_{ac,\min}$ ensures adequate harmonic suppression and safe short-circuit current.
- *DC-side inductance L_{dc}:* The upper bound $L_{dc,\max}$ is limited by required dynamic response, size, and cost, while the lower bound $L_{dc,\min}$ ensures DC ripple stays within specifications to prevent equipment aging and process quality degradation.

The truncated normal distribution enforces physical bounds while maintaining mathematical tractability for Monte Carlo propagation.

2) Monte Carlo simulation: MCS approach is adopted, by which a set of 5000 random sample pairs (L_{ac}, L_{dc}) is generated using rejection sampling to meet the defined constraints. For each of the parameter samples, a HSS model is built and solved. The iterative process involves:

- Recalculating the commutation angle μ based on the sampled pair of parameters.
- Reconstructing time-periodic switching functions $S^i_{dc,ac}(t)$, $S_{dc}(t)$, and $S^{i,j}_{ac}(t)$ based on the updated commutation angle.
- Numerically solving the HSS equations to extract and store the magnitudes and phase angles of the AC-side harmonic currents of interest.

C. Dependency Analysis and Probability Density Function Fitting

A case study is conducted based on a representative 6-pulse LCC unit, with nominal capacity of 1.25 MW and nominal line voltage of 380 V, which is connected to the 35 kV bus through a rectifier transformer with a ratio of 35 kV/400 V. For the AC-side commutation inductance, the truncation bounds are $L_{ac,\min} = 0.01$ mH and $L_{ac,\max} = 0.03$ mH, corresponding to transformer short-circuit voltage percentages ranging from $U_k = 5\%$ to 15%. For the DC-side smoothing inductance, the truncation bounds are $L_{dc,\min} = 0.05$ mH and $L_{dc,\max} = 0.35$ mH.

Fig.2 shows the harmonic phasor distribution of each characteristic harmonic, caused by parameter uncertainty. In each harmonic phasor distribution plot, the red one represents the phasor with mean magnitude and mean phase angle, while the blue ones represent the phasors within the 95% confidence intervals (CI). The results indicate that: 1) the harmonic phasors at different frequencies exhibit mutual dependency; 2) higher orders of harmonics display progressively larger phaser angle variations and uncertainties.

1) Inter-Harmonic Dependency Analysis: An important feature of the harmonics generated by LCC is the inter-harmonic dependency due to frequency coupling as indicated by the HHS model in Equations (3). To assess this dependency from the probabilistic distributions obtained via MCS, the correlation structure between different harmonic components is investigated. Correlation matrices, with Kendall's Tau correlation coefficients, are used to characterize the dependencies.

As shown in Fig. 3, the phase angle correlation matrix $\mathbf{R}_{\text{phase}}$ exhibits very strong correlations with the Gaussian copula parameters $\rho^{Gauss}_{ij} \geq 0.97$, indicating distinct characteristics in the inter-harmonic dependency structure for phase angles.

2) Marginal Distribution Fitting: The marginal distributions of each harmonic component are modeled independently. The best-fit PDF for each harmonic magnitude and phase angle is fitted to the data obtained from MCS. The candidate distributions considered include Normal, Lognormal, Beta, Logistic and Exponential distributions.

For each harmonic h, the marginal distribution $F_h(x_h)$ is estimated by fitting multiple parametric distributions and selecting the best-fit distribution based on minimizing the Root Mean Squared Error (RMSE) between the cumulative distribution functions simulated and fitted. The best-fit results are illustrated in Fig.4, showing the histograms and the fitted PDFs of harmonic magnitudes and phase angles.

Tables II summarize the best-fit distributions and their estimated parameters for harmonic phase angles.

TABLE II
BEST-FIT DISTRIBUTIONS FOR HARMONIC PHASE ANGLES

Order	PDF	RMSE	Parameters
1st	Beta	$9.30e^{-3}$	$[10.8052, 20.2376, -107.9443, 13.0245]$
5th	Beta	$1.97e^{-3}$	$[11.8144, 23.6725, -2.6825, 70.8689]$
7th	Beta	$1.54e^{-3}$	$[9.7918, 18.8775, -29.9737, 88.2789]$
11th	Beta	$7.41e^{-4}$	$[12.4463, 22.1493, 59.5585, 164.3101]$
13th	Beta	$6.84e^{-4}$	$[10.4088, 17.2887, 29.4667, 172.4882]$

The parameters of the Beta distribution are given as $[\alpha, \beta, \ell, \theta]$, where α and β are the shape parameters, and ℓ and θ are the location and scale parameters, respectively.

3) Shared Beta Distribution for Phase Angles: Since strong inter-harmonic dependencies are proved for phase angles, a shared Beta distribution is used to model the phase angle distributions of all harmonic jointly. In this way, only two shape parameters need to be estimated for all harmonics, while

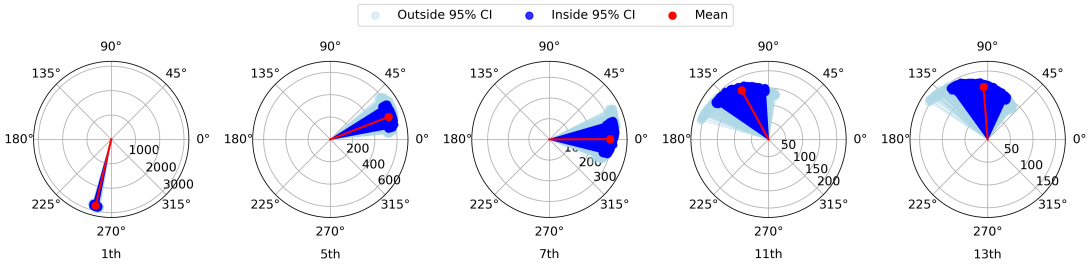


Fig. 2. Harmonic phasor distributions caused by parameter uncertainty

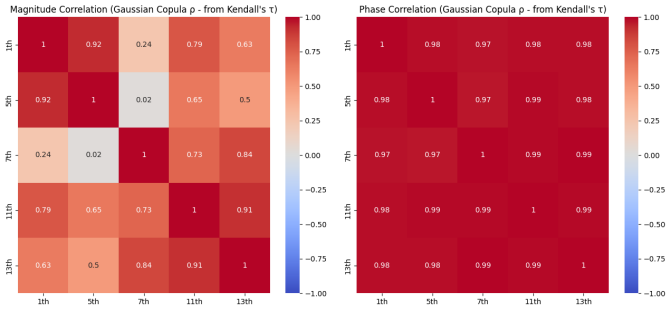


Fig. 3. Gaussian copula correlation matrices for harmonic magnitudes and harmonic phase angles.

each harmonic retains its own location and scale parameters to capture the specific range of phase angles.

Firstly, a normalization procedure is applied to each harmonic's unwrepped phase angle data to map them onto the unit interval $[0, 1]$. Let $\phi_h = \{\phi_{h,i}\}_{i=1}^{N}$ denote the unwrapped phase samples in degrees for harmonic $h \in \{1, 5, 7, 11, 13\}$, where unwrapping is performed to avoid discontinuities at $\pm 180°$. For each harmonic, the observed min–max values are represented as:

$$v_{\min,h} = \min(\phi_h), \qquad v_{\max,h} = \max(\phi_h). \tag{6}$$

The sampled data of each harmonic phase angle is then normalized to the unit interval via the min-max transformation:

$$u_{h,i} = \frac{\phi_{h,i} - v_{\min,h}}{v_{\max,h} - v_{\min,h}} \in (0, 1). \tag{7}$$

Secondly, all normalized samples from the five harmonics are pooled into a single dataset:

$$\mathcal{U} = \bigcup_{h \in \{1,5,7,11,13\}} \{u_{h,i}\}_{i=1}^{N}. \tag{8}$$

A single Beta distribution with estimated shape parameters $\hat{\alpha}$ and $\hat{\beta}$ is fitted to $\mathcal{U}$ via maximum likelihood estimation, with location and scale fixed to the standard interval $[0, 1]$:

$$u \sim \text{Beta}(\hat{\alpha}, \hat{\beta}). \tag{9}$$

The result of the fitted parameters are $\hat{\alpha} = 4.0414$, $\hat{\beta} = 4.7762$.

The Beta probability density function is given by:

$$f_{\text{Beta}}(u; \hat{\alpha}, \hat{\beta}) = \frac{u^{\hat{\alpha}-1}(1-u)^{\hat{\beta}-1}}{\int_0^1 t^{\hat{\alpha}-1}(1-t)^{\hat{\beta}-1}\, dt}, \quad u \in (0, 1). \tag{10}$$

Thirdly, the fitted shared Beta distribution is transformed back to each harmonic's sampled phase angle scale to assess the goodness-of-fit. For each harmonic h, the denormalization is performed as:

$$f_{\phi_h}(\phi) = \frac{1}{v_{\max,h} - v_{\min,h}} f_{\text{Beta}}\left(\frac{\phi - v_{\min,h}}{v_{\max,h} - v_{\min,h}}; \hat{\alpha}, \hat{\beta}\right), \tag{11}$$

where $\phi \in [v_{\min,h}, v_{\max,h}]$ is the original phase angle range for harmonic h.

Table III lists the denormalized parameters (location $v_{\min,h}$ and scale $v_{\max,h} - v_{\min,h}$) for each harmonic. These parameters define the linear mapping from the standard $[0, 1]$ interval back to the sampled phase ranges. The shared shape parameters $\hat{\alpha}$ and $\hat{\beta}$ remain constant across all harmonics.

The goodness-of-fit of the shared Beta distribution model is evaluated for each harmonic phase angle by comparing the denormalized Beta PDF against the empirical histograms from MCS. Root Mean Squared Error (RMSE) is computed between the empirical histogram and the denormalized Beta PDF, with the results summarized in last column of Table III.

The comparison plots between the empirical histograms and the denormalized shared Beta PDFs for each harmonic phase angle are shown in Fig. 5. Both the visual comparison and the low RMSE values indicate a good fit of the shared Beta distribution to the phase angle data across all harmonics.

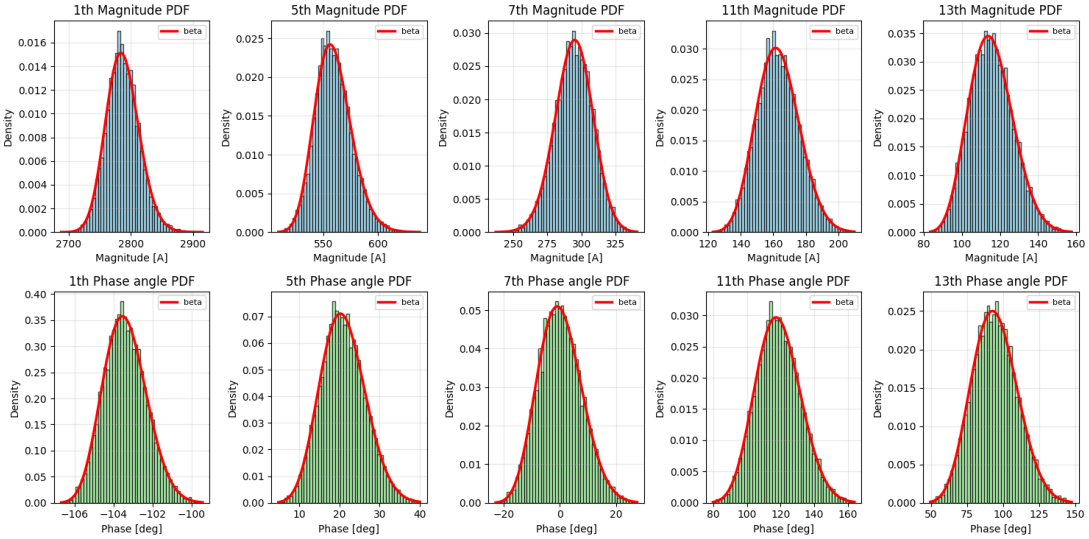


Fig. 4. Harmonic phasor distributions with corresponding best-fit probability density functions.

TABLE III
SHARED BETA DISTRIBUTION PARAMETERS AND FIT QUALITY FOR HARMONIC PHASE ANGLES

Harmonic	$\hat{\alpha}$	$\hat{\beta}$	$v_{\min}$ [deg]	$v_{\max}$ [deg]	$v_{\max} - v_{\min}$ [deg]	RMSE
1st	4.041	4.776	-106.70	-99.43	7.27	0.0226
5th	4.041	4.776	4.75	40.05	35.30	0.0037
7th	4.041	4.776	-22.77	27.51	50.28	0.0029
11th	4.041	4.776	79.69	164.01	84.32	0.0014
13th	4.041	4.776	49.38	147.67	98.29	0.0011

Note: All harmonics share the same shape parameters ($\hat{\alpha} = 4.041$, $\hat{\beta} = 4.776$).
Location parameter $v_{\min}$ and scale parameter $(v_{\max} - v_{\min})$ are harmonic-specific.

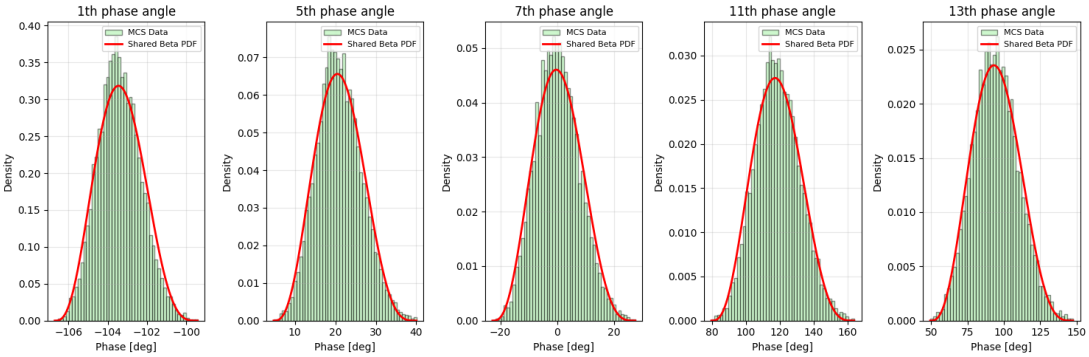


Fig. 5. Empirical histograms and denormalized shared Beta PDFs for harmonic phase angles.

III. Mathematical Framework

This section presents the mathematical framework for the stochastic harmonic hosting capacity problem of a phase-balanced three-phase power system, considering uncertainty on the angle of the harmonic current injected by the connected harmonic units. To this end, the formulation for the deterministic harmonic hosting capacity presented in [1] is extended. For brevity, the original formulation is not repeated in this section, rather the relevant variables and constraints are highlighted and transformed to their stochastic counterpart.

First, Section III-A introduces the generalized polynomial chaos expansion method used to incorporate random variables into the formulation. Second, Section III-B introduces the uncertainty on the angle of the injected harmonic current and a method to approximate the related trigonometric functions in the context of polynomial chaos expansion. Third, Section III-C presents the transformation of the deterministic formulation into its stochastic counterpart and characterizes the resulting problem class. Finally, Section III-D shows that for the absolute equality fairness criterion the problem reduces to an exact linear program. For the full stochastic formulation of the harmonic hosting capacity problem, the reader is referred to Appendix A.

In the following table, a general overview and generic examples of the notation used throughout the mathematical framework are provided.

element	notation	example
index	lower-case italic	x
set	calligraphic	$\mathcal{X}$
(random) variable	sans serif	X
parameter	italic	X
approximation	hat notation	$\hat{\mathsf{X}}$
vector	bar notation	$\bar{\mathsf{X}}$, $\bar{X}$

A. Background on Polynomial Chaos Expansion

This section introduces the necessary background on polynomial chaos expansion, a method for representing random variables in terms of polynomial functions. Concretely, a vector of input random variables, e.g., harmonic current angles, is propagated through the set of constraints of the harmonic hosting capacity problem to obtain a vector of unknown output random variables, e.g., harmonic bus voltages.

First, uncertainty quantification of random variables is introduced in terms of polynomial chaos expansion. Second and third, the impact of the polynomial chaos expansion method on equality and inequality constraints is discussed, respectively. For a detailed treatment of the subject, the reader is referred to [2].

1) Uncertainty Quantification: In general, it is assumed that the stochastic input is captured by a real-valued finite-variance stochastic germ $\bar{\omega} = [\omega_1, ..., \omega_N]^{\mathrm{T}}$. Optimization solvers are unable to handle equations involving random variables directly, necessitating the formulation of a deterministic equivalent of the stochastic problem. For this purpose, polynomial chaos expansion approximates any real-valued random variable x of finite variance that is a function of the stochastic germ $\bar{\omega}$ as a linear combination $\hat{\mathsf{x}}$ of the orthogonal polynomial basis $[\psi_k(\bar{\omega})]_{k\in\mathcal{K}}$, where $\mathcal{K} \subseteq \mathbb{N}_0$, as,

$$\hat{\mathsf{x}} = \sum_{k\in\mathcal{K}} x_k \cdot \psi_k = \sum_{k\in\mathcal{K}} \frac{\langle \mathsf{x}, \psi_k \rangle}{\langle \psi_k, \psi_k \rangle} \cdot \psi_k, \tag{12}$$

where $x_k \in \mathbb{R}$ denote the polynomial chaos expansion coefficients and $\langle \cdot, \cdot \rangle$ denotes the inner product. Through the use of the polynomial chaos expansion coefficients, a deterministic equivalent can be built for the stochastic problem. The choice of orthogonal polynomial basis depends on the distribution of the stochastic germ. In this work, following the Askey scheme [2], Beta-distributed germ corresponds to Jacobi polynomials.

2) Uncertainty Projection onto Equality Constraints: An optimization problem requires an intrusive approach to exploit polynomial chaos expansion. Through polynomial chaos expansion, any stochastic equality constraint is equivalently represented by $|\mathcal{K}|$ deterministic equality constraints. The total number of resulting equations, known as the cardinality of the model $|\mathcal{K}|$, is determined by the number of independent uncertainty sources and the highest polynomial degree as,

$$|\mathcal{K}| = \frac{(|\bar{\omega}| + \delta)!}{|\bar{\omega}|!\, \delta!}, \tag{13}$$

where δ denotes the maximum degree of the polynomial basis functions. The truncation error $\|\mathsf{x} - \hat{\mathsf{x}}\|$ decays to zero for $|\mathcal{K}| \to \infty$.

Additionally, the Galerkin projection is required to form a system of equations for the stochastic modes, which are generally coupled together (Ch. 12 in [2]). For the purpose of the harmonic hosting capacity problem, in terms of the Galerkin projection, two distinct operations stand out: summation and multiplication. As an example, consider three approximated random variables $\hat{\mathsf{x}}$, $\hat{\mathsf{y}}$ and $\hat{\mathsf{z}}$ defined using the same polynomial basis. The Galerkin projection (gp) of both operations is,

$$\hat{\mathsf{z}} = \hat{\mathsf{x}} + \hat{\mathsf{y}} \overset{\text{gp}}{\to} z_k = x_k + y_k, \qquad \forall k \in \mathcal{K}, \tag{14}$$

$$\hat{\mathsf{z}} = \hat{\mathsf{x}} \cdot \hat{\mathsf{y}} \overset{\text{gp}}{\to} z_k = \sum_{\bar{\kappa}\in P^*(\mathcal{K},2)} M_{\bar{\kappa},k}\,(x_{\bar{\kappa}_1} \cdot y_{\bar{\kappa}_2}), \quad \forall k \in \mathcal{K}, \tag{15}$$

where $P^*(\mathcal{K}, 2)$ denotes the 2-tuples of $\mathcal{K}$. The multiplication tensor $M_{\bar{\kappa},k}$ is determined as,

$$M_{\bar{\kappa},k} = \frac{\langle \psi_{\bar{\kappa}_1}, ..., \psi_{\bar{\kappa}_{|\bar{\kappa}|}}, \psi_k \rangle}{\langle \psi_k, \psi_k \rangle}, \tag{16}$$

and can be computed ahead-of-time. For $k = 0$, orthogonality of the basis implies $M_{\bar{\kappa},0}$ is diagonal with strictly positive entries; for $k \neq 0$, the tensor has mixed-sign off-diagonal entries, and the resulting matrix $[M_{\bar{\kappa},k}]_{\bar{\kappa}}$ is generally indefinite.

3) Chance-Constraint Reformulation of Inequality Constraints: All harmonic limits in this work are upper bounds. It is infeasible to enforce a deterministic upper bound on an infinite-support continuous random variable. Alternatively, a chance constraint ensures that the probability of x exceeding a deterministic limit $x^{\max}$ is below a risk level ε:

$$\mathbb{P}(\mathsf{x} \leq x^{\max}) \geq 1 - \varepsilon. \tag{17}$$

Since the output random variables are nonlinear functions of a Beta-distributed input, their output distribution is generally

non-Gaussian and unknown. The distribution-free one-sided Cantelli bound is therefore adopted,

$$\mathbb{E}(\mathsf{x}) + \lambda(\varepsilon)\sqrt{\mathbb{V}(\mathsf{x})} \leq x^{\max}, \quad \lambda(\varepsilon) = \sqrt{\frac{1-\varepsilon}{\varepsilon}}, \quad (18)$$

which guarantees the chance constraint for any distribution with finite variance [2], without any assumption on how uncertainty propagates through the nonlinear network equations.

The expectation $\mathbb{E}$ and variance $\mathbb{V}$ of an approximated random variable $\hat{\mathsf{x}}$ are expressed in terms of the polynomial chaos expansion coefficients as,

$$\mathbb{E}[\hat{\mathsf{x}}] = x_0, \quad (19)$$

$$\mathbb{V}[\hat{\mathsf{x}}] = \sum_{k \in \mathcal{K} \setminus \{0\}} \langle \psi_k, \psi_k \rangle \, x_k^2. \quad (20)$$

Substituting (19)–(20) into (18) gives the deterministic equivalent in terms of the polynomial chaos expansion coefficients, which is a second-order cone constraint in the coefficients $\{x_k\}_{k \in \mathcal{K}}$.

B. Uncertainty on Angle of Harmonic Current Injection

This section discusses the treatment of uncertain angles $\phi_{u,h}$ of harmonic currents injected by harmonic units $u \in \mathcal{U}^{\mathrm{h}}$, for all harmonics $h \in \mathcal{H}$. Throughout, a polynomial basis of degree $\delta = 2$ is used, as it has been shown that polynomial basis functions with degree two capture the non-linear nature of the power flow equations without significant loss of accuracy [**?**].

As shown in Section II, the phase angles of a given unit across different harmonic orders exhibit strong mutual dependency, and their marginal distributions are well approximated by a single shared Beta distribution with shared shape parameters and harmonic-specific location and scale parameters. Each harmonic angle is recovered using a harmonic-specific affine transformation. The angles of different units are assumed to be mutually independent, each driven by its own Beta distribution, so the total germ vector has dimension $|\mathcal{U}^{\mathrm{h}}|$ and the cardinality of the model is $|\mathcal{K}| = (|\mathcal{U}^{\mathrm{h}}| + 2)!/(|\mathcal{U}^{\mathrm{h}}|!\, 2!)$. Given the affine transformation, the polynomial chaos expansion coefficients of $\phi_{u,h}$ are non-zero only at indices $k = 0$ and $k = 1$ within the $\delta = 2$ basis.

In the deterministic formulation of the harmonic hosting capacity problem presented in [1], the harmonic current magnitude $\mathsf{I}^{\mathrm{mag}}_{u,h}$ is related to its rectangular components using the angle $\phi_{u,h}$, as,

$$\cos(\phi_{u,h})\, \mathsf{I}^{\mathrm{mag}}_{u,h} = \mathsf{I}^{\mathrm{re}}_{u,h}, \quad \forall u \in \mathcal{U}^{\mathrm{h}}, h \in \mathcal{H}, \quad (21)$$

$$\sin(\phi_{u,h})\, \mathsf{I}^{\mathrm{mag}}_{u,h} = \mathsf{I}^{\mathrm{im}}_{u,h}, \quad \forall u \in \mathcal{U}^{\mathrm{h}}, h \in \mathcal{H}. \quad (22)$$

Direct substitution of the approximated angle $\hat{\phi}_{u,h}$ is impossible, since the Galerkin projection is not defined for trigonometric functions. Instead, as suggested in [2], the trigonometric functions are approximated by a truncated Taylor series,

$$\cos(\phi) \approx \hat{\cos}(\phi) = 1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!} - \frac{\phi^6}{6!} + \dots, \quad (23)$$

$$\sin(\phi) \approx \hat{\sin}(\phi) = \phi - \frac{\phi^3}{3!} + \frac{\phi^5}{5!} - \frac{\phi^7}{7!} + \dots, \quad (24)$$

whose Galerkin projection is well-defined. Exploiting (15) and the known polynomial chaos expansion coefficients of $\hat{\phi}_{u,h}$, the Galerkin projection yields precomputed scalar coefficients $\cos(\phi_{u,h})_k$ and $\sin(\phi_{u,h})_k$ for each $k \in \mathcal{K}$. It is important to note that these are parameters, not optimization variables. A four-term Taylor series ($n = 4$) is adequate for first-quadrant Beta-distributed angles. Finally, exploitation of the trigonometric addition relations,

$$\sin(\phi + \varphi) = \sin(\phi)\cos(\varphi) + \cos(\phi)\sin(\varphi), \quad (25)$$

$$\cos(\phi + \varphi) = \cos(\phi)\cos(\varphi) - \sin(\phi)\sin(\varphi), \quad (26)$$

allows any uncertain angle outside the first quadrant to be written as the sum of a first-quadrant uncertain angle ϕ and a deterministic offset φ. Since φ is deterministic, the Galerkin projection of (25)–(26) reduces to a linear combination via (14).

C. General Stochastic Harmonic Hosting Capacity

This section presents the reformulation of the deterministic harmonic hosting capacity model presented in [1] to obtain a stochastic harmonic hosting capacity model using polynomial chaos expansion. The stochastic model aims at determining a *deterministic* harmonic hosting capacity given uncertainty on the harmonic current angles, while complying with chance constraints on (harmonic) voltages and currents. Given that the problem is formulated from the perspective of the power system operator, it is most practical to suggest deterministic rather than stochastic harmonic budgets for its customers, as interpretation of the latter becomes obscure. A deterministic harmonic hosting capacity in a polynomial chaos expansion context implies that $\mathsf{I}^{\mathrm{mag}}_{u,h,k} = 0$ for any non-zero index k. As such, the polynomial chaos expansion counterparts of (21) and (22) are,

$$\cos(\phi_{u,h})_k\, \mathsf{I}^{\mathrm{mag}}_{u,h,0} = \mathsf{I}^{\mathrm{re}}_{u,h,k}, \quad \forall u \in \mathcal{U}^{\mathrm{h}}, h \in \mathcal{H}, k \in \mathcal{K}, \quad (27)$$

$$\sin(\phi_{u,h})_k\, \mathsf{I}^{\mathrm{mag}}_{u,h,0} = \mathsf{I}^{\mathrm{im}}_{u,h,k}, \quad \forall u \in \mathcal{U}^{\mathrm{h}}, h \in \mathcal{H}, k \in \mathcal{K}. \quad (28)$$

Since $\cos(\phi_{u,h})_k$ and $\sin(\phi_{u,h})_k$ are precomputed parameters, constraints (27)–(28) are *linear* in the decision variables $\mathsf{I}^{\mathrm{mag}}_{u,h,0}$, $\mathsf{I}^{\mathrm{re}}_{u,h,k}$, and $\mathsf{I}^{\mathrm{im}}_{u,h,k}$. Furthermore, any variables related to the chosen fairness objective are also treated as deterministic, i.e., only non-zero for index $k = 0$. All other parameters, e.g., network impedances, are deterministic.

In the remainder of this section, the necessary constraint reformulations are presented, classified as §III-C1 linear constraints, or §III-C2 quadratic constraints. First, the applicable constraints in the deterministic model are enumerated, and second, the reformulation is introduced in a general sense and is exemplified.

1) Reformulation of Linear Constraints: Constraints (10)–(31), (34), (36), and (38)–(41) in the deterministic harmonic hosting capacity model [1] are all linear. Consider a generic linear constraint,

$$a\,\mathsf{x} + b\,\mathsf{y} = c, \quad (29)$$

where the parameters a, b and c are deterministic, and variables x and y are stochastic. Supported by (14), the reformulation of a linear constraint in a polynomial chaos expansion

is straightforward, i.e., repeat the original constraint over all indices $k \in \mathcal{K}$:

$$a\,\mathsf{x}_k + b\,\mathsf{y}_k = c_k, \quad \forall k \in \mathcal{K}, \tag{30}$$

where $c_k = c$ for $k = 0$ and $c_k = 0$ for $k \neq 0$ when c is a deterministic constant. The resulting set of $|\mathcal{K}|$ equality constraints is linear in the PCE coefficients of the decision variables.

2) Reformulation of Quadratic Constraints and Problem Class: Constraints (47)–(49) and (51) in the deterministic harmonic hosting capacity model [1] take the form of second-order cone constraints. Consider a generic second-order cone constraint,

$$\mathsf{x}^2 + \mathsf{y}^2 \leq a^2, \tag{31}$$

where the parameter a is deterministic and the variables x, y are stochastic. Following the strategy of [3], an auxiliary lifted variable z is introduced as,

$$\mathsf{x}^2 + \mathsf{y}^2 = \mathsf{z} \leq a^2. \tag{32}$$

Supported by (15), in a polynomial chaos expansion context, the left-hand side of (32) yields the following $|\mathcal{K}|$ constraints on the PCE coefficients of the auxiliary variable z,

$$\sum_{\bar{\kappa} \in P^*(\mathcal{K},2)} M_{\bar{\kappa},k} \left(\mathsf{x}_{\kappa_1} \cdot \mathsf{x}_{\kappa_2} + \mathsf{y}_{\kappa_1} \cdot \mathsf{y}_{\kappa_2}\right) = \mathsf{z}_k, \quad \forall k \in \mathcal{K}. \tag{33}$$

These constraints are quadratic in the PCE coefficients of x and y. Their convexity depends on the index k. On the one hand, for $k = 0$, orthogonality of the polynomial basis renders $M_{\bar{\kappa},0}$ diagonal with positive entries, making the quadratic form positive semidefinite and the constraint convex. On the other hand, for $k \neq 0$, the tensor $M_{\bar{\kappa},k}$ has mixed-sign off-diagonal entries, making the quadratic form indefinite and the constraint non-convex. Consequently, the stochastic harmonic hosting capacity problem is a non-convex quadratically constrained quadratic program. Consequently, the treatment of (33) differs by index. For $k = 0$, the equality is relaxed to an inequality,

$$\sum_{\bar{\kappa} \in P^*(\mathcal{K},2)} M_{\bar{\kappa},0} \left(\mathsf{x}_{\kappa_1} \cdot \mathsf{x}_{\kappa_2} + \mathsf{y}_{\kappa_1} \cdot \mathsf{y}_{\kappa_2}\right) \leq \mathsf{z}_0, \tag{34}$$

which is a convex rotated second-order cone constraint. This relaxation is exact at optimality: since z_0 is the expectation of z and appears with a positive coefficient in the chance constraint (36), the optimizer minimises it and will always set $\mathsf{z}_0 = Q_0(\mathsf{x}, \mathsf{y})$ at any optimal solution, so (34) binds at equality. This tightness is verified numerically in the first case study of Section IV. For $k \neq 0$, the equality in (33) must be retained,

$$\sum_{\bar{\kappa} \in P^*(\mathcal{K},2)} M_{\bar{\kappa},k} \left(\mathsf{x}_{\kappa_1} \cdot \mathsf{x}_{\kappa_2} + \mathsf{y}_{\kappa_1} \cdot \mathsf{y}_{\kappa_2}\right) = \mathsf{z}_k, \quad \forall k \in \mathcal{K}\backslash\{0\}. \tag{35}$$

The reason is that relaxing (33) to an inequality for $k \neq 0$ would introduce a correctness error in the chance constraint. Since the quadratic form $Q_k(\mathsf{x}, \mathsf{y})$ is indefinite, it can take negative values. An inequality $Q_k \leq \mathsf{z}_k$ would allow the optimizer to set $\mathsf{z}_k = 0$ whenever $Q_k < 0$, as this is both feasible and minimises the variance term $\langle \psi_k, \psi_k \rangle\, \mathsf{z}_k^2$ in the chance constraint. However, the true PCE coefficient is $\mathsf{z}_k = Q_k < 0$, with a variance contribution $\langle \psi_k, \psi_k \rangle\, Q_k^2 > 0$. Setting $\mathsf{z}_k = 0$ would therefore underestimate the variance and produce an over-optimistic chance constraint. Enforcing equality (35) ensures that z_k always tracks the true k-th PCE coefficient of $\mathsf{x}^2 + \mathsf{y}^2$, regardless of its sign.

Using the auxiliary variables $\{\mathsf{z}_k\}_{k \in \mathcal{K}}$ and (**??**), the right-hand side of (32) is reformulated as a Cantelli chance constraint,

$$\mathsf{z}_0 + \sqrt{\frac{1-\varepsilon}{\varepsilon}} \sqrt{\sum_{k \in \mathcal{K}\backslash\{0\}} \langle \psi_k, \psi_k \rangle\, \mathsf{z}_k^2} \leq a^2. \tag{36}$$

This constraint is a second-order cone in the auxiliary variables $\{\mathsf{z}_k\}_{k \in \mathcal{K}}$: it is convex, and does not contribute to the non-convexity of the problem.

D. Exact Linear Program for Absolute Equality

When the absolute equality fairness criterion is applied, all harmonic units share a single hosting capacity per harmonic,

$$\mathsf{I}^{\text{mag}}_{u,h,0} = I_h, \quad \forall u \in \mathcal{U}^{\text{h}}, h \in \mathcal{H}, \tag{37}$$

so that the vector of hosting capacities collapses to a scalar I_h per harmonic. Under this condition, together with a single harmonic unit per bus or when all units share the same injection point, the non-convexity of the lifting constraints disappears entirely, and the problem reduces to an exact linear program, as shown below.

1) Reduction to Scalar Decision Variables: Since the IV-formulation network equations (KCL, Ohm's law, transformer constraints) are all linear in the PCE coefficients, and since the harmonic current injection constraints (27)–(28) express each PCE coefficient of the injected current as a known parameter times I_h, all voltage and branch current PCE coefficients are affine functions of the scalar I_h,

$$\mathsf{U}^{\text{re}}_{i,h,k} = C^{\text{re}}_{i,h,k} \cdot I_h, \qquad \mathsf{U}^{\text{im}}_{i,h,k} = C^{\text{im}}_{i,h,k} \cdot I_h, \tag{38}$$

where $C^{\text{re}}_{i,h,k}$ and $C^{\text{im}}_{i,h,k}$ are precomputed constants obtained from the harmonic admittance matrix and the trigonometric PCE coefficients. An analogous relation holds for branch and transformer currents.

2) Collapse of Quadratic Lifting to Scalar Form: Substituting (38) into the lifting constraint (**??**) for voltage, the quadratic form reduces to,

$$I_h^2 \cdot \alpha_{i,h,k} \leq \mathsf{W}_{i,h,k}, \quad \forall i \in \mathcal{I}, h \in \mathcal{H}, k \in \mathcal{K}, \tag{39}$$

where the scalar,

$$\alpha_{i,h,k} = \sum_{\bar{\kappa} \in P^*(\mathcal{K},2)} M_{\bar{\kappa},k} \Big(C^{\text{re}}_{i,h,\bar{\kappa}_1} C^{\text{re}}_{i,h,\bar{\kappa}_2} + C^{\text{im}}_{i,h,\bar{\kappa}_1} C^{\text{im}}_{i,h,\bar{\kappa}_2} \Big) \tag{40}$$

is a precomputed parameter. Constraint (39) is a convex quadratic in I_h regardless of the sign of $\alpha_{i,h,k}$: if $\alpha_{i,h,k} \leq 0$, the constraint is satisfied trivially for all $I_h \geq 0$; if $\alpha_{i,h,k} > 0$, it is a convex upper bound on I_h^2.

3) Reduction of Chance Constraints to Linear Bounds: Substituting (39) at equality into the chance constraint (36), and collecting all precomputed terms, each chance constraint collapses to a single linear bound on I_h,

$$I_h^2 \cdot C_{i,h} \leq a_{i,h}^2, \quad \Longleftrightarrow \quad I_h \leq \frac{a_{i,h}}{\sqrt{C_{i,h}}}, \tag{41}$$

where $a_{i,h}$ is the deterministic limit (e.g., the IHD voltage limit) and the composite scalar,

$$C_{i,h} = \alpha_{i,h,0} + \sqrt{\frac{1-\varepsilon}{\varepsilon}} \sqrt{\sum_{k \in \mathcal{K} \setminus \{0\}} \langle \psi_k, \psi_k \rangle \, \alpha_{i,h,k}^2}, \tag{42}$$

is precomputed from network data and PCE parameters. Since $\alpha_{i,h,0} \geq 0$ and the second term is non-negative, $C_{i,h} \geq 0$ always, and the bound (41) is valid and finite.

This result holds for all chance constraints in the model: IHD voltage, THD voltage, RMS voltage, and RMS current all reduce to a bound of the form (41), with the appropriate limit $a_{i,h}$ and composite scalar $C_{i,h}$. The stochastic harmonic hosting capacity problem under absolute equality therefore reduces to,

$$\begin{aligned} &\max_{I_h} \sum_{h \in \mathcal{H}} |\mathcal{U}^{\mathrm{h}}| \cdot I_h, \\ &\text{s.t.} \quad I_h \leq \frac{a_{i,h,c}}{\sqrt{C_{i,h,c}}}, \quad \forall i \in \mathcal{I}, h \in \mathcal{H}, c \in \mathcal{C}, \end{aligned} \tag{43}$$

where $\mathcal{C}$ indexes the set of active chance constraints (IHD, THD, RMS voltage, RMS current). This is an *exact linear program*: each harmonic hosting capacity is bounded by the tightest applicable constraint, and all precomputed constants absorb both the network topology and the stochastic angle uncertainty. The optimal solution is obtained in closed form as,

$$I_h^* = \min_{i,\,c} \frac{a_{i,h,c}}{\sqrt{C_{i,h,c}}}. \tag{44}$$

IV. Numerical Illustration

This section presents the numerical illustration of the stochastic harmonic hosting capacity problem of a phase-balanced three-phase power system, considering harmonic recombination. To this end, first, a five bus section of a real-world industrial power system is considered to show the equivalence between the (33) and (**??**). Second, a large-scale transmission system is considered, to show the scalability of the proposed second-order cone formulation. For the purpose of these numerical illustrations, any resistive impedance is scaled with a factor $\sqrt{h}$, and any inductive or capacitive impedance is appropriately scaled with a factor h. All results are presented in per-unit using a 100 MVA basis.

A. Industrial Power System

B. Large-scale Transmission System

V. Conclusion

APPENDIX

APPENDIX A
OPTIMIZATION MODEL

1) the fundamental harmonic 1 is excluded from the set $\mathcal{H}$,
2) absolute equality fairness criteria is used as the harmonic hosting capacity objective,
3) the index 0 for the harmonic current magnitude $\mathsf{I}^{\text{mag}}_{u,h,0}$ for the harmonic unit is purely decorative as the variable is not defined for any other polynomial chaos expansion index

$$\max \sum_{u\in\mathcal{U}^{\text{h}}} \sum_{h\in\mathcal{H}} \mathsf{I}^{\text{mag}}_{u,h,0}, \tag{45}$$

$$\text{s.t. } \mathsf{I}^{\text{mag}}_{u,h,0} = \mathsf{I}^{\text{mag}}_{v,h,0},\ \forall u,v\in\mathcal{U}^{\text{h}}, h\in\mathcal{H}, \tag{46}$$

BUS CONSTRAINTS

reference bus voltage – $\forall i\in\mathcal{I}^{\text{ref}}, h\in\mathcal{H}$:

$$\mathsf{U}^{\text{re}}_{i,h,0} = U^{\text{ref,re}}_{i,h}, \tag{47}$$

$$\mathsf{U}^{\text{im}}_{i,h,0} = U^{\text{ref,im}}_{i,h}, \tag{48}$$

Kirchhoff's current law – $\forall i\in\mathcal{I}, h\in\mathcal{H}, k\in\mathcal{K}$:

$$\sum_{eij\in\mathcal{T}^{\text{e}}} \mathsf{I}^{\text{re}}_{eij,h,k} + \sum_{ui\in\mathcal{T}^{\text{u}}} \mathsf{I}^{\text{re}}_{ui,h,k} + g^{\text{sh}}_{i,h}\mathsf{U}^{\text{re}}_{i,h,k} - b^{\text{sh}}_{i,h}\mathsf{U}^{\text{im}}_{i,h,k} = 0, \tag{49}$$

$$\sum_{eij\in\mathcal{T}^{\text{e}}} \mathsf{I}^{\text{im}}_{eij,h,k} + \sum_{ui\in\mathcal{T}^{\text{u}}} \mathsf{I}^{\text{im}}_{ui,h,k} + g^{\text{sh}}_{i,h}\mathsf{U}^{\text{im}}_{i,h,k} + b^{\text{sh}}_{i,h}\mathsf{U}^{\text{re}}_{i,h,k} = 0, \tag{50}$$

BRANCH CONSTRAINTS

Ohm's law – $\forall bij\in\mathcal{T}^{\text{b}\rightarrow}, h\in\mathcal{H}, k\in\mathcal{K}$:

$$\mathsf{U}^{\text{re}}_{j,h,k} = \mathsf{U}^{\text{re}}_{i,h,k} - r^{\text{s}}_{b,h}\mathsf{I}^{\text{s,re}}_{bij,h,k} + x^{\text{s}}_{b,h}\mathsf{I}^{\text{s,im}}_{bij,h,k}, \tag{51}$$

$$\mathsf{U}^{\text{im}}_{j,h,k} = \mathsf{U}^{\text{im}}_{i,h,k} - r^{\text{s}}_{b,h}\mathsf{I}^{\text{s,im}}_{bij,h,k} - x^{\text{s}}_{b,h}\mathsf{I}^{\text{s,re}}_{bij,h,k}, \tag{52}$$

branch total current – $\forall bij\in\mathcal{T}^{\text{b}}, h\in\mathcal{H}, k\in\mathcal{K}$:

$$\mathsf{I}^{\text{re}}_{bij,h,k} = g^{\text{sh}}_{bij,h}\mathsf{U}^{\text{re}}_{i,h,k} - b^{\text{sh}}_{bij,h}\mathsf{U}^{\text{im}}_{i,h,k} + \mathsf{I}^{\text{s,re}}_{bij,h,k}, \tag{53}$$

$$\mathsf{I}^{\text{im}}_{bij,h,k} = g^{\text{sh}}_{bij,h}\mathsf{U}^{\text{im}}_{i,h,k} + b^{\text{sh}}_{bij,h}\mathsf{U}^{\text{re}}_{i,h,k} + \mathsf{I}^{\text{s,im}}_{bij,h,k}, \tag{54}$$

TRANSFORMER CORE CONSTRAINTS

Ohm's law – $\forall xij\in\mathcal{T}^{\text{x}\rightarrow}, h\in\mathcal{H}, k\in\mathcal{K}$:

$$\mathsf{V}^{\text{re}}_{xij,h,k} = \mathsf{E}^{\text{re}}_{x,h,k} - x^{\text{s}}_{x,h}\,\mathsf{I}^{\text{s,im}}_{xij,h,k}, \tag{55}$$

$$\mathsf{V}^{\text{im}}_{xij,h,k} = \mathsf{E}^{\text{im}}_{x,h,k} + x^{\text{s}}_{x,h}\,\mathsf{I}^{\text{s,re}}_{xij,h,k}, \tag{56}$$

Kirchhoff's current law – $\forall xij\in\mathcal{T}^{\text{x}\rightarrow}, h\in\mathcal{H}, k\in\mathcal{K}$:

$$\mathsf{I}^{\text{s,re}}_{xji,h,k} + t^{\text{vg,re}}_{x,h}\left(\mathsf{I}^{\text{s,re}}_{xij,h,k} - \mathsf{I}^{\text{m,re}}_{x,h,k} - \frac{\mathsf{E}^{\text{re}}_{x,h,k}}{r^{\text{sh}}_{x,h}}\right) + t^{\text{vg,im}}_{x,h}\left(\mathsf{I}^{\text{s,im}}_{xij,h,k} - \mathsf{I}^{\text{m,im}}_{x,h,k} - \frac{\mathsf{E}^{\text{im}}_{x,h,k}}{r^{\text{sh}}_{x,h}}\right) = 0 \tag{57}$$

$$\mathsf{I}^{\text{s,im}}_{xji,h,k} + t^{\text{vg,re}}_{x,h}\left(\mathsf{I}^{\text{s,im}}_{xij,h,k} - \mathsf{I}^{\text{m,im}}_{x,h,k} - \frac{\mathsf{E}^{\text{im}}_{x,h,k}}{r^{\text{sh}}_{x,h}}\right) + t^{\text{vg,im}}_{x,h}\left(\mathsf{I}^{\text{s,re}}_{xij,h,k} - \mathsf{I}^{\text{m,re}}_{x,h,k} - \frac{\mathsf{E}^{\text{re}}_{x,h,k}}{r^{\text{sh}}_{x,h}}\right) = 0 \tag{58}$$

transformer phase shift – $\forall xji\in\mathcal{T}^{\text{x}\leftarrow}, h\in\mathcal{H}, k\in\mathcal{K}$:

$$\mathsf{E}^{\text{re}}_{x,h,k} = t^{\text{vg,re}}_{x,h}\,\mathsf{V}^{\text{re}}_{xji,h,k} - t^{\text{vg,im}}_{x,h}\,\mathsf{V}^{\text{im}}_{xji,h,k}, \tag{59}$$

$$\mathsf{E}^{\text{im}}_{x,h,k} = t^{\text{vg,re}}_{x,h}\,\mathsf{V}^{\text{im}}_{xji,h,k} + t^{\text{vg,im}}_{x,h}\,\mathsf{V}^{\text{re}}_{xji,h,k},$$

TRANSFORMER WINDING CONSTRAINTS

Kirchhoff's current law – $\forall xij \in \mathcal{T}^{\text{x}}, h \in \mathcal{H}, k \in \mathcal{K}$:

$$\mathsf{I}^{\text{s,re}}_{xij,h,k} = \mathsf{I}^{\text{re}}_{xij,h,k} + g^{\text{sh}}_{xij,h}\,\mathsf{V}^{\text{re}}_{xij,h,k} - b^{\text{sh}}_{xij,h}\,\mathsf{V}^{\text{im}}_{xij,h,k},$$

$$\mathsf{I}^{\text{s,im}}_{xij,h,k} = \mathsf{I}^{\text{im}}_{xij,h,k} + g^{\text{sh}}_{xij,h}\,\mathsf{V}^{\text{im}}_{xij,h,k} + b^{\text{sh}}_{xij,h}\,\mathsf{V}^{\text{re}}_{xij,h,k},$$

Ohm's law – $\forall xij \in \mathcal{T}^{\text{x}}, h \in \mathcal{H}^{+} \cup \mathcal{H}^{-}, k \in \mathcal{K}$:

$$\mathsf{U}^{\text{re}}_{i,h,k} = \mathsf{V}^{\text{re}}_{xij,h,k} + r_{xij,h}\,\mathsf{I}^{\text{re}}_{xij,h,k},$$

$$\mathsf{U}^{\text{im}}_{i,h,k} = \mathsf{V}^{\text{im}}_{xij,h,k} + r_{xij,h}\,\mathsf{I}^{\text{im}}_{xij,h,k},$$

Ohm's law – $\forall xij \in \mathcal{T}^{\text{x,earthed}}, h \in \mathcal{H}^{0}, k \in \mathcal{K}$:

$$\mathsf{U}^{\text{re}}_{i,h,k} = \mathsf{V}^{\text{re}}_{xij,h,k} + (r_{xij,h} + 3r^{0}_{xij,h})\mathsf{I}^{\text{re}}_{xij,h,k} - 3x^{0}_{xij,h}\mathsf{I}^{\text{im}}_{xij,h,k},$$

$$\mathsf{U}^{\text{im}}_{i,h,k} = \mathsf{V}^{\text{im}}_{xij,h,k} + (r_{xij,h} + 3r^{0}_{xij,h})\mathsf{I}^{\text{im}}_{xij,h,k} + 3x^{0}_{xij,h}\mathsf{I}^{\text{re}}_{xij,h,k},$$

cur. blocking – $\forall xij \in \mathcal{T}^{\text{x,ne}} \cup \mathcal{T}^{\text{x,delta}}, h \in \mathcal{H}^{0}, k \in \mathcal{K}$:

$$\mathsf{I}^{\text{re}}_{xij,h,k} = 0,$$

$$\mathsf{I}^{\text{im}}_{xij,h,k} = 0.$$

harmonic current injection – $\forall u \in \mathcal{U}^{\text{h}}, h \in \mathcal{H}, k \in \mathcal{K}$:

$$\cos(\phi_{u,h})_k\,\mathsf{I}^{\text{mag}}_{u,h,0} = \mathsf{I}^{\text{re}}_{u,h,k},$$

$$\sin(\phi_{u,h})_k\,\mathsf{I}^{\text{mag}}_{u,h,0} = \mathsf{I}^{\text{im}}_{u,h,k},$$

auxiliary variables – $\forall i \in \mathcal{I}, h \in \mathcal{H}, k \in \mathcal{K}$:

$$\sum_{\bar{\kappa} \in P^{*}(\mathcal{K},2)} M_{\bar{\kappa},k}\left(\mathsf{I}^{\text{re}}_{i,h,\bar{\kappa}_1} \cdot \mathsf{I}^{\text{re}}_{i,h,\bar{\kappa}_2} + \mathsf{I}^{\text{im}}_{i,h,\bar{\kappa}_1} \cdot \mathsf{I}^{\text{im}}_{i,h,\bar{\kappa}_2}\right) \leq \mathsf{J}_{i,h,k}, \quad (71)$$

$$\sum_{\bar{\kappa} \in P^{*}(\mathcal{K},2)} M_{\bar{\kappa},k}\left(\mathsf{U}^{\text{re}}_{i,h,\bar{\kappa}_1} \cdot \mathsf{U}^{\text{re}}_{i,h,\bar{\kappa}_2} + \mathsf{U}^{\text{im}}_{i,h,\bar{\kappa}_1} \cdot \mathsf{U}^{\text{im}}_{i,h,\bar{\kappa}_2}\right) \leq \mathsf{W}_{i,h,k},$$

root mean square voltage – $\forall i \in \mathcal{I}$:

$$\mathbb{E}(\sum_{h \in \mathcal{H}} \bar{\mathsf{W}}_{i,h}) + \lambda(\varepsilon)\sqrt{\mathbb{V}(\sum_{h \in \mathcal{H}} \bar{\mathsf{W}}_{i,h})} \leq (U^{\text{max}}_{i})^2 - (U^{\text{mag,fund}}_{i})^2,$$

total harmonic voltage – $\forall i \in \mathcal{I}$:

$$\mathbb{E}(\sum_{h \in \mathcal{H}} \bar{\mathsf{W}}_{i,h}) + \lambda(\varepsilon)\sqrt{\mathbb{V}(\sum_{h \in \mathcal{H}} \bar{\mathsf{W}}_{i,h})} \leq (U^{\text{thd}}_{i})^2\,(U^{\text{mag,fund}}_{i})^2,$$

individual harmonic voltage – $\forall i \in \mathcal{I}, h \in \mathcal{H}$:

$$\mathbb{E}(\bar{\mathsf{W}}_{i,h}) + \lambda(\varepsilon)\sqrt{\mathbb{V}(\bar{\mathsf{W}}_{i,h})} \leq (U^{\text{ihd}}_{i,h})^2\,(U^{\text{mag,fund}}_{i})^2.$$

root mean square current – $\forall eij \in \mathcal{T}^{\text{e}}$:

$$\mathbb{E}(\sum_{h \in \mathcal{H}} \bar{\mathsf{J}}_{i,h}) + \lambda(\varepsilon)\sqrt{\mathbb{V}(\sum_{h \in \mathcal{H}} \bar{\mathsf{J}}_{i,h})} \leq (I^{\text{rated}}_{eij})^2 - (I^{\text{mag,fund}}_{eij})^2, \quad (76)$$

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